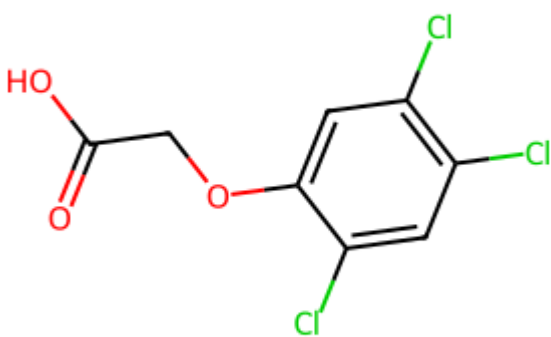


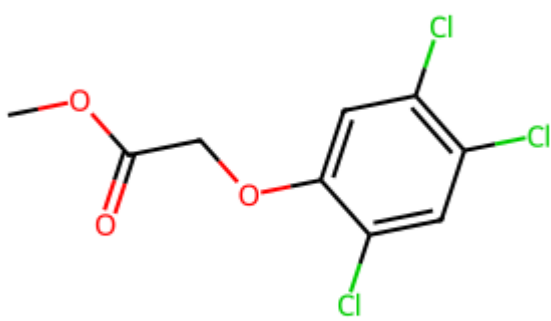


```
{'analysis_time': 0.0,  
'first_solution_time': 0.13302,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 8.782550223499053,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 8.782550223499053,  
'retro_star_value': 8.782550223499053}
```



Reaction

```
{'analysis_time': 0.13302,  
'first_solution_time': 0.319879,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 8.782550223499053,  
'retro_star_reaction_number': 8.782550223499053,  
'retro_star_rxn_cost': 1.0278089142358144,  
'retro_star_value': 8.782550223499053}
```

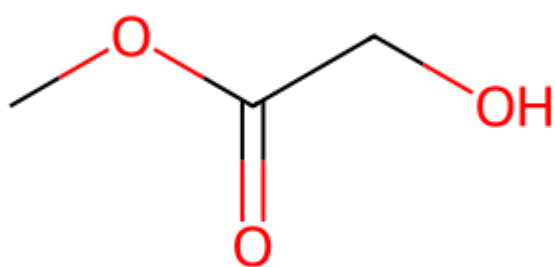
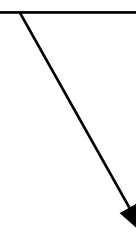
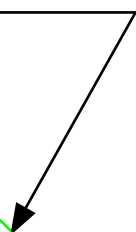


```
{'analysis_time': 0.13302,  
'first_solution_time': 0.319879,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 7.754741309263238,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 7.754741309263238,  
'retro_star_value': 8.782550223499053}
```

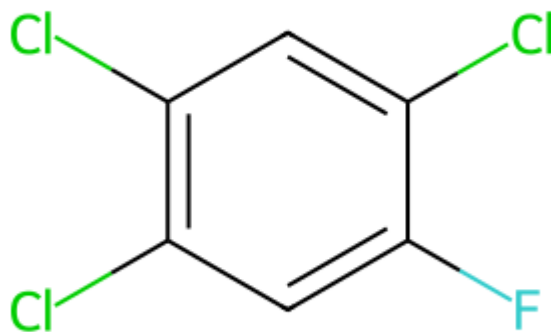


Reaction

```
{'analysis_time': 0.319879,  
'first_solution_time': 0.319879,  
'num_calls_rxn_model': 3,  
'num_calls_value_function': 70,  
'retro_star_min_cost': 10.372684681131167,  
'retro_star_reaction_number': 10.372684681131167,  
'retro_star_rxn_cost': 10.372684681131167,  
'retro_star_value': 11.400493595366981}
```



```
{'analysis_time': 0.319879,  
'first_solution_time': 0.319879,  
'num_calls_rxn_model': 3,  
'num_calls_value_function': 70,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 11.400493595366981}
```



```
{'analysis_time': 0.319879,  
'first_solution_time': 0.319879,  
'num_calls_rxn_model': 3,  
'num_calls_value_function': 70,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 11.400493595366981}
```